ASSIGNMENT – C02AT03

NAME: M. GOKUL
REG NO:19256067

COURSE: DEVSECOPS

CODE: CSA5803

CASE STUDY

TITLE: Prepare a secure workflow for an airport passenger management system with threat analysis and architectural security controls.

Abstract

Airports process millions of passengers every day, making passenger management systems one of the most critical infrastructures in transportation. These systems handle sensitive information such as passenger identities, passports, biometric data, flight schedules, and baggage details. Due to increasing cyber threats such as data breaches, ransomware, insider attacks, and identity theft, a secure workflow is essential to protect passenger information and ensure uninterrupted airport operations.

This case study presents a secure airport passenger management workflow, identifies major security threats at each stage, and proposes architectural security controls to mitigate these risks. The proposed architecture follows the principles of **Confidentiality, Integrity, and Availability (CIA)** while implementing authentication, encryption, access control, intrusion detection, network segmentation, and continuous monitoring.

1. Introduction

An Airport Passenger Management System (APMS) is an integrated platform that manages:

- Passenger registration
- Ticket verification
- Identity validation
- Security screening
- Immigration clearance
- Boarding process
- Flight information

- Baggage tracking

As airports increasingly adopt digital technologies, cyberattacks have become a significant concern. Therefore, implementing a secure architecture is necessary to protect passenger privacy and maintain operational continuity.

2. Objectives

- Design a secure passenger workflow.
- Identify possible cyber threats.
- Apply security controls to each workflow stage.
- Protect passenger data from unauthorized access.
- Ensure secure communication between airport systems.

Background

Sky Secure International Airport is one of the busiest airports, serving more than 60,000 passengers daily. The airport uses a centralized Airport Passenger Management System (APMS) to manage ticket booking, passenger check-in, baggage handling, immigration verification, boarding, and flight information.

The system stores confidential passenger data, including passport information, biometric records, travel history, payment details, and boarding information. Because of the large amount of sensitive information processed, the airport has become a target for cybercriminals attempting phishing attacks, ransomware, insider threats, malware infections, and database attacks.

To strengthen cybersecurity, the airport decides to redesign its passenger management system using a **Zero Trust Security Architecture** with multiple layers of protection. The new system secures every stage of the passenger journey through strong authentication, encrypted communication, continuous monitoring, secure APIs, endpoint protection, and advanced threat detection.

This case study analyzes the secure workflow of the Airport Passenger Management System, identifies major cyber threats, and recommends architectural security controls to ensure safe, reliable, and uninterrupted airport operations.

Aim

The aim of this study is to design and analyze a secure workflow for an Airport Passenger Management System by identifying potential cyber threats, evaluating system vulnerabilities, and implementing architectural security controls to protect passenger information, ensure secure communication, maintain system availability, and support safe and efficient airport operations.

Objectives

- To understand the working of an Airport Passenger Management System.
- To design a secure workflow for passenger processing.
- To identify cyber threats affecting airport systems.
- To perform threat analysis using the STRIDE model.
- To implement architectural security controls.
- To protect passenger information using encryption and access control.
- To improve system availability and resilience against cyberattacks.
- To recommend cybersecurity best practices for airport infrastructure.

Theory

An **Airport Passenger Management System (APMS)** is an integrated software platform that manages the complete lifecycle of a passenger, from ticket reservation to aircraft boarding. It enables airlines and airport authorities to automate operations such as online booking, check-in, baggage tracking, immigration verification, boarding, and flight scheduling. The system improves operational efficiency while ensuring accurate passenger records and regulatory compliance.

Since APMS handles confidential information such as passenger names, passport details, biometric data, travel history, and payment information, protecting the system against cyber threats is essential. A successful cyberattack may result in data breaches, identity theft, financial fraud, operational disruption, or even threats to national security.

The security of an Airport Passenger Management System is based on the **Confidentiality, Integrity, and Availability (CIA) Triad**:

- **Confidentiality** ensures that only authorized users can access passenger information through encryption, authentication, and access control.
- **Integrity** ensures that stored and transmitted information remains accurate and protected against unauthorized modification using digital signatures, hashing, and audit logs.
- **Availability** guarantees uninterrupted access to airport services through load balancing, backups, disaster recovery, and redundant infrastructure.

Modern airports also adopt a **Zero Trust Security Model**, which assumes that no user or device should be trusted by default. Every access request must be authenticated, authorized, and continuously verified before access is granted.

To secure the Airport Passenger Management System, organizations implement multiple architectural security controls, including:

- **Transport Layer Security (TLS 1.3)** for secure communication.
- **Multi-Factor Authentication (MFA)** for user verification.
- **Role-Based Access Control (RBAC)** to limit user permissions.
- **Web Application Firewall (WAF)** to protect against web attacks.
- **API Gateway** to secure communication between applications.
- **Intrusion Detection and Prevention Systems (IDS/IPS)** to detect malicious network activities.
- **Endpoint Detection and Response (EDR)** to monitor and protect endpoint devices.
- **Security Information and Event Management (SIEM)** for continuous monitoring and incident response.

- **AES-256 encryption** for protecting passenger databases and sensitive information.
- **Backup and Disaster Recovery** solutions to ensure business continuity in the event of cyberattacks or system failures.

By combining these security mechanisms with continuous monitoring, vulnerability management, regular software updates, and employee cybersecurity awareness, airports can maintain secure operations while protecting passenger privacy and critical infrastructure. This layered security approach significantly reduces cyber risks and ensures the confidentiality, integrity, and availability of airport passenger management services.

1. System Components of Airport Passenger Management System

a) Passenger Registration Module

- Stores passenger details such as name, passport number, contact information, and travel preferences.
- Performs identity validation before creating passenger profiles.
- Uses encryption to protect personally identifiable information (PII).

Security Controls:

- Data encryption
- Input validation
- Secure database access
- Identity verification

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b) Ticket Reservation and Booking Module

This module allows passengers to:

- Search available flights
- Reserve seats

- Make payments
- Receive booking confirmations

Security Risks:

- Payment fraud
- Account takeover
- Fake booking requests

Security Controls:

- Secure payment gateway
- Token-based authentication
- Fraud detection system
- Transaction monitoring

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c) Check-in Management Module

The check-in system supports:

- Online check-in
- Airport counter check-in
- Self-service kiosk check-in

Security Features:

- QR code-based boarding passes
- Identity verification
- Secure session management
- Anti-tampering mechanisms

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d) Baggage Management System

The baggage module tracks passenger luggage from check-in to aircraft loading.

Functions:

- Generate baggage tags
- Track baggage location
- Prevent baggage loss
- Monitor suspicious activities

Security Controls:

- RFID tracking
- Secure API communication
- Access restrictions
- Real-time monitoring

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e) Immigration and Biometric Verification

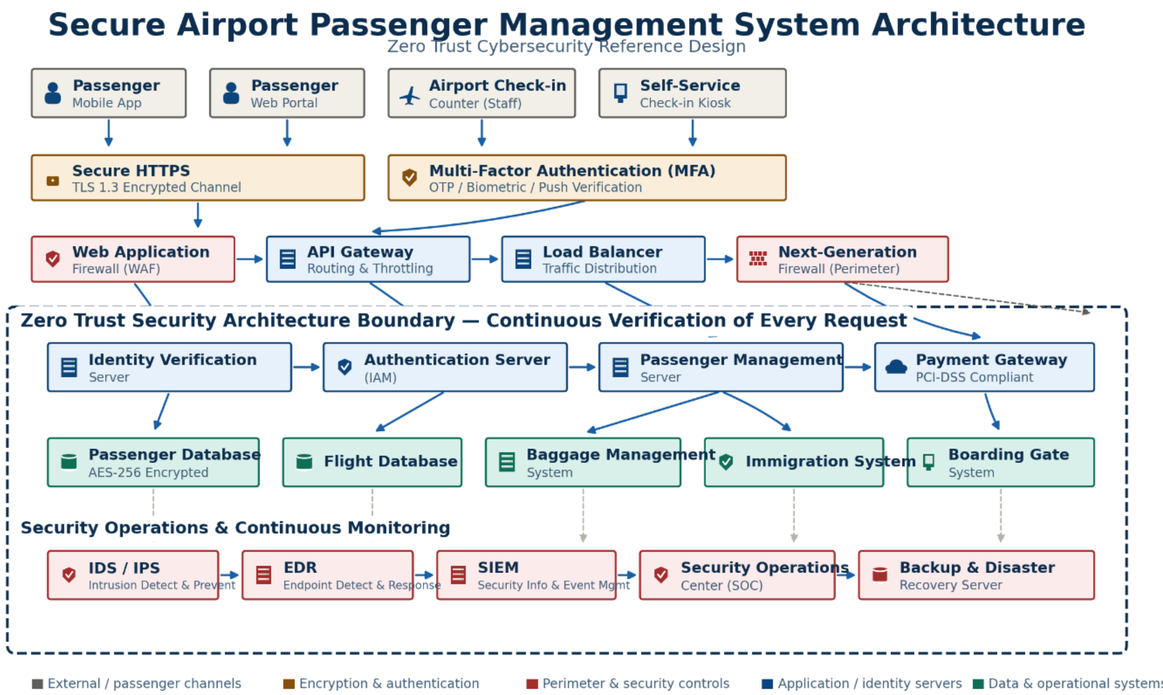
This module verifies passenger identity using:

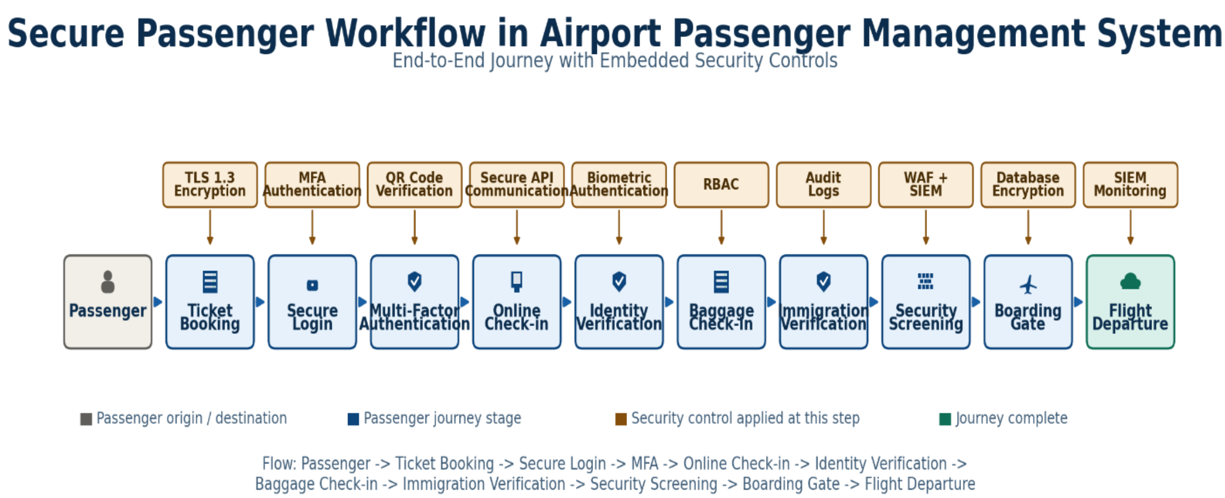
- Passport scanning
- Facial recognition
- Fingerprint verification

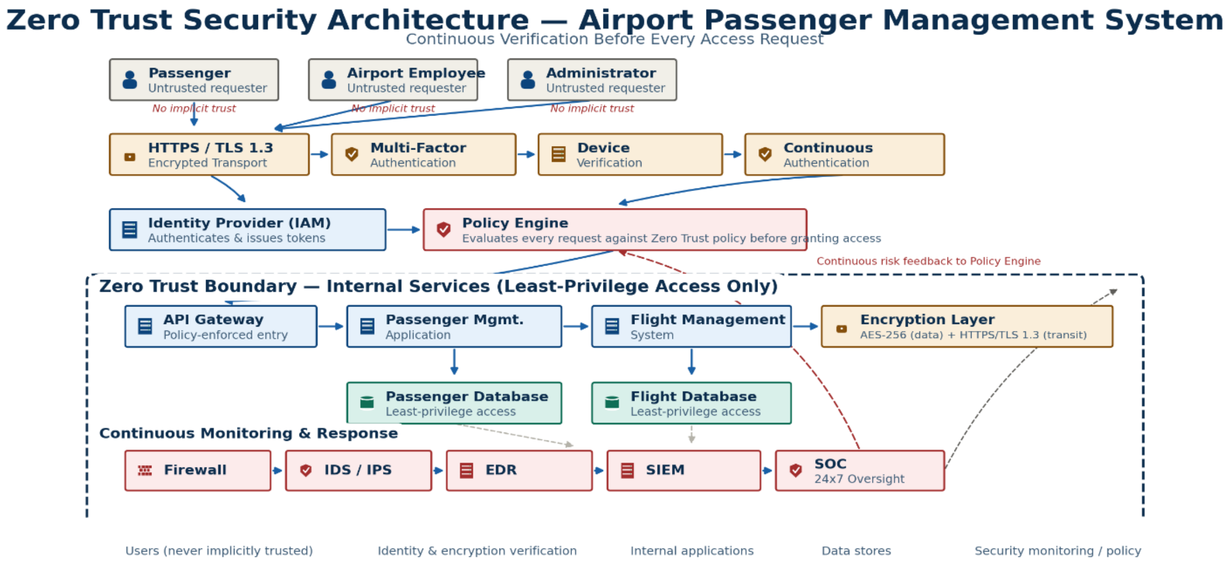
Security Controls:

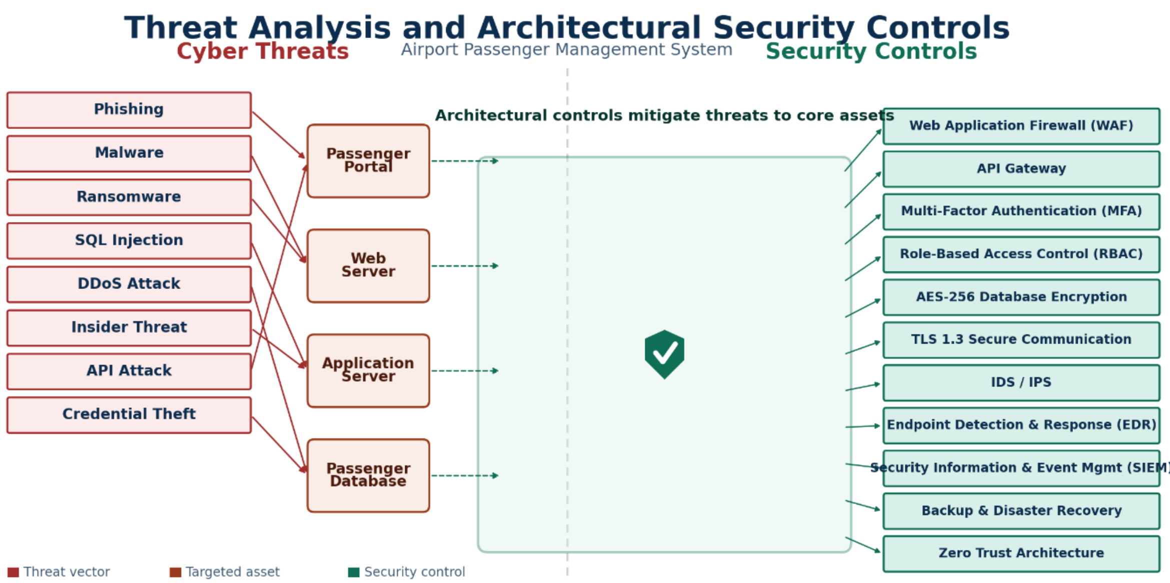
- Biometric encryption
- Secure identity database
- Access logging
- Privacy protection

ARCHITECTURE DIAGRAM :






Zero Trust Security Architecture — Airport Passenger Management System
Continuous Verification Before Every Access Request
Passenger
Untrusted requester
Airport Employee
Untrusted requester
Administrator
Untrusted requester
No implicit trust
No implicit trust
No implicit trust
HTTPS / TLS 1.3
Encrypted Transport
Multi-Factor
Authentication
Device
Verification
Continuous
Authentication
Identity Provider (IAM)
Authenticates & issues tokens
Policy Engine
Evaluates every request against Zero Trust policy before granting access
Continuous risk feedback to Policy Engine
Zero Trust Boundary — Internal Services (Least-Privilege Access Only)
API Gateway
Policy-enforced entry
Passenger Mgmt.
Application
Flight Management
System
Encryption Layer
AES-256 (data) + HTTPS/TLS 1.3 (transit)
Passenger Database
Least-privilege access
Flight Database
Least-privilege access
Continuous Monitoring & Response
Firewall
IDS / IPS
EDR
SIEM
SOC
24x7 Oversight
Users (never implicitly trusted)
Identity & encryption verification
Internal applications
Data stores
Security monitoring / policy


Threat Analysis and Architectural Security Controls
Cyber Threats
Airport Passenger Management System
Security Controls
Phishing
Malware
Ransomware
SQL Injection
DDoS Attack
Insider Threat
API Attack
Credential Theft
Passenger Portal
Web Server
Application Server
Passenger Database
Architectural controls mitigate threats to core assets
Web Application Firewall (WAF)
API Gateway
Multi-Factor Authentication (MFA)
Role-Based Access Control (RBAC)
AES-256 Database Encryption
TLS 1.3 Secure Communication
IDS / IPS
Endpoint Detection & Response (EDR)
Security Information & Event Mgmt (SIEM)
Backup & Disaster Recovery
Zero Trust Architecture
Threat vector
Targeted asset
Security control